

UNIT-2

Structured Analysis

- Structured Analysis is a development method that allows the analyst to understand the system and its activities in a logical way.
- It is a systematic approach, which uses graphical tools that analyze and refine the objectives of an existing system and develop a new system specification which can be easily understandable by user.
- It is a set of techniques and graphical tools that allow the analyst to develop a new kind of system that is understandable to the user .
- It is a development method that allows the analyst to understand the system and its activities in a logical way

It has following attributes –

- It is graphic which specifies the presentation of application.
- It divides the processes so that it gives a clear picture of system flow.
- It is logical rather than physical i.e., the elements of system do not depend on vendor or hardware.
- It is an approach that works from high-level overviews to lower-level details.

Steps involved in structured analysis process

- I. Planning the approach
- II. Asking questions and collecting data
- III. Recording the information

Why do we use these tools? :

- Use of graphics whenever possible to help communicate better with the user.
- Differentiate between logical and physical system.
- Build a logical system model to familiarize the user with system characteristics and interrelationships before implementation.

Major steps in Structured Analysis

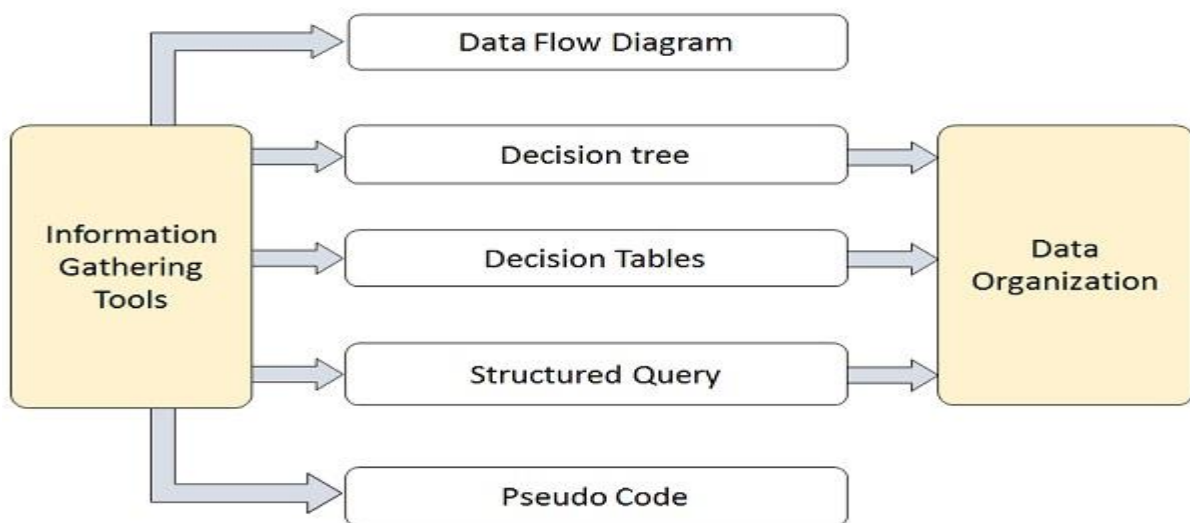
- Studying the current business environment
- Modeling the old logical system
- Modeling a new logical system

- Modeling a new physical environment
- Evaluating alternatives
- Selecting the best design
- Creating structured

Structured Analysis tools:

During Structured Analysis, various tools and techniques are used for system development. They are:

1. Data Flow Diagrams
2. Data Dictionary
3. Decision Trees
4. Decision Tables
5. Structured English
6. Gantt chart



DFD:-

1. DFD is stand for data flow diagram.
2. DFD is also known as bubble chart.
3. Data flow diagram is graphical representation of flow of data in an information system.
4. It is capable of depicting (to show) incoming data flow, outgoing data flow and stored data.
5. DFDs depict flow of data in the system at various levels

6. DFD does not contain any control or branch elements
7. Data Flow diagrams are very popular because they help us to visualize the major steps and data involved in software-system processes.

Symbols/ components/ notation Used in DFD

1. **Square Box:** A square box defines source or destination of the system. It is also called entity. It is represented by rectangle.
2. **Arrow or Line:** An arrow identifies the data flow i.e. it gives information to the data that is in motion.
3. **Circle or bubble chart:** It represents as a process that gives us information. It is also called processing box.
4. **Open Rectangle:** An open rectangle is a data store. In this data is store either temporary or permanently.

Symbol Name	Symbol	Meaning
Square		Source or Destination of Data
Arrow		Data flow
Circle		Process transforming data flow
Open Rectangle		Data Store

Levels of DFD

DFD uses hierarchy to maintain transparency thus multilevel DFD's can be created. Levels of DFD are as follows:

1. 0-level DFD:

1. It represents the entire system as a single bubble and provides an overall picture of the system.
2. It represents the entire system as a single bubble with input and output data indicated by incoming/outgoing arrows.



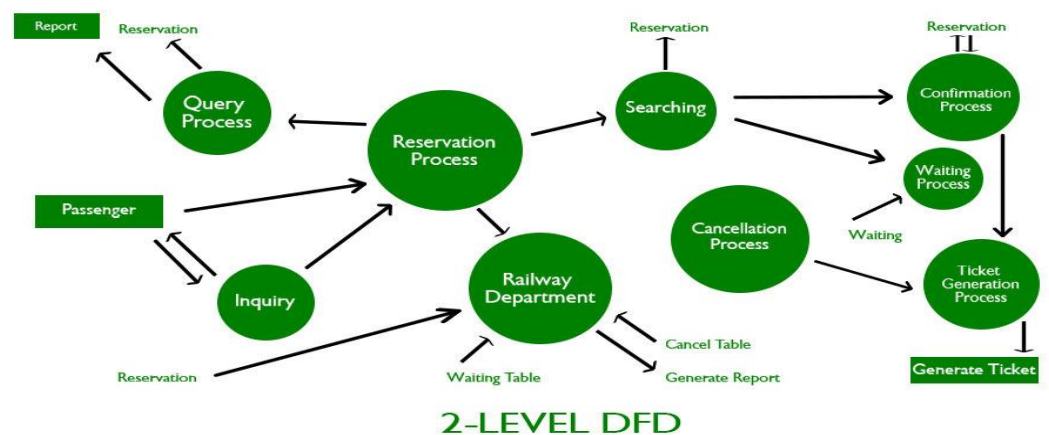
2. 1-level DFD:

1. It represents the main functions of the system and how they interact with each other.
2. The Level 0 DFD is broken down into more specific, known as Level 1 DFD.
3. Level 1 DFD also mentions basic processes and sources of information.



3. 2-level DFD:

1. It represents the processes within each function of the system and how they interact with each other.
2. 2-Level DFD goes one step deeper into parts of 1-level DFD.
3. It can be used to plan or record the specific/necessary detail about the system's functioning.



Advantages of DFD

- It helps us to understand the functioning and the limits of a system.
- It is a graphical representation which is very easy to understand as it helps visualize contents.
- Data Flow Diagram represent detailed and well explained diagram of system components.
- It is used as the part of system documentation file.
- Data Flow Diagrams can be understood by both technical or nontechnical person because they are very easy to understand.

Disadvantages of DFD

- At times DFD can confuse the programmers regarding the system.
- Data Flow Diagram takes long time to be generated, and many times due to this reasons analysts are denied permission to work on it.

Types of DFD:-

DFDs are of two types: Physical DFD and Logical DFD. The following table lists the points that differentiate a physical DFD from a logical DFD.

Physical DFD	Logical DFD
It is implementation dependent. It shows which functions are performed.	It is implementation independent. It focuses only on the flow of data between processes.
It provides low level details of hardware, software, files, and people.	It explains events of systems and data required by each event.
It depicts how the current system operates and how a system will be implemented.	It shows how business operates; not how the system can be implemented.

Data Dictionary:-

- A data dictionary is a collection of the names, definitions, and attributes for data elements and models.
- The data in a data dictionary is the metadata about the database. { metadata i.e data about the database)
- A data dictionary is a structured repository of data elements in the system.
- It stores the descriptions of all DFD data elements that is, details and definitions of data flows, data stores, data stored in data stores, and the processes.
- A data dictionary improves the communication between the analyst and the user.
- It plays an important role in building a database. Most DBMSs have a data dictionary as a standard feature. For example, refer the following table –

Sr.No.	Data Name	Description	No. of Characters
1	ISBN	ISBN Number	10
2	TITLE	Title	60
3	SUB	Book Subjects	80
4	ANAME	Author Name	15

Purpose and importance of a data dictionary

- A data dictionary can help team members learn everything about a data set.

- The data dictionary will define conventions for the project and consistency throughout the dataset.
- The main reason companies use data dictionaries is to document and share data structures and other information for all involved with a project or database

Gantt Chart:-

- It is one of the most popular and useful ways of showing activities (tasks or events) displayed against time.
- Each activity is represented by a bar.
- the position and length of the bar reflects the start date, duration and end date of the activity.
- the chart is displayed as a horizontal bar chart.
- Horizontal bars of different lengths represent the project timeline, which can include task sequences, duration, and the start and end dates for each task. The horizontal bar also shows how much of a task requires completion.
- A Gantt chart helps in scheduling, managing, and monitoring specific tasks and resources in a project.
- The chart shows the project timeline, which includes scheduled and completed work over a period.

Gantt charts used for :

- Monitor project progress
- Adjust project schedule
- Optimize the complete process structure
- Estimate delays
- Keep all project participants on track
- Monitoring progress to ensure completion on schedule

Gantt Chart

Task Name	Q1 2019			Q2 2019		Q3 2019
	Jan 19	Feb 19	Mar 19	Apr 19	Jun 19	Jul 19
Planning						
Research						
Design						
Implementation						
Follow up						

Flowchart

- It is a pictorial representation of Algorithm for planning the procedure to solve a program/problem.
- A flowchart is a picture of the separate steps of a process in sequential order.
- It is a generic tool that can be adapted for a wide variety of purposes,
- It can be used to describe various processes, such as a manufacturing process, an administrative or service process, or a project plan.

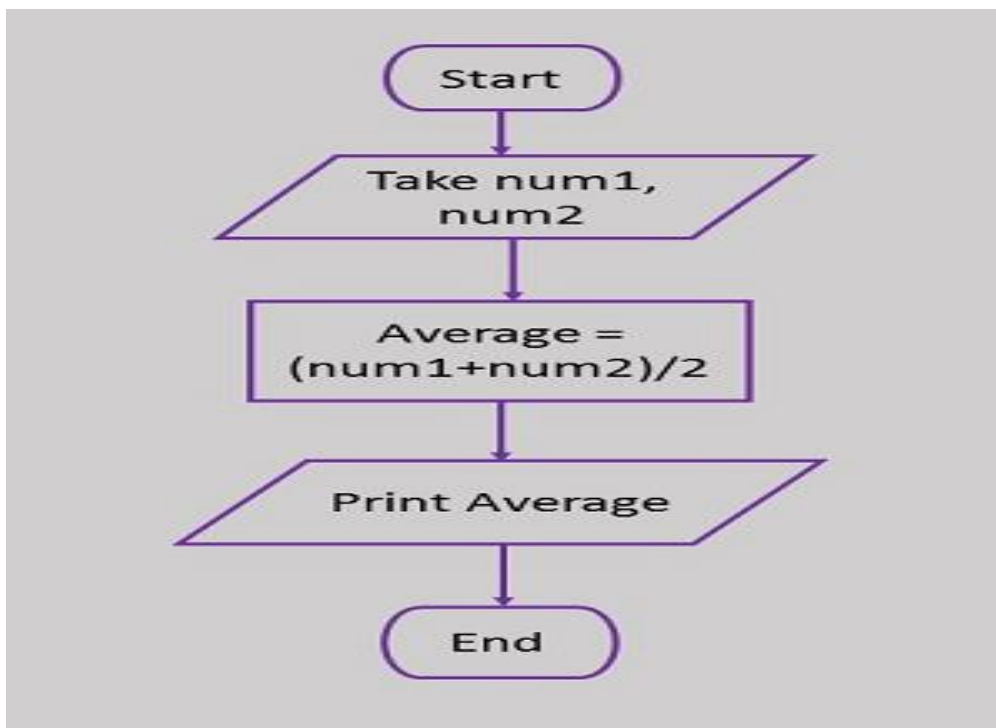
Flowchart symbol:-

Symbol	Name	Function
	Start/end	An oval represents a start or end point
	Arrows	A line is a connector that shows relationships between the representative shapes
	Input/Output	A parallelogram represents input or output
	Process	A rectangle represents a process
	Decision	A diamond indicates a decision

Guidelines for Developing Flowcharts

These are some points to keep in mind while developing a flowchart –

1. Flowchart can have only one start and one stop symbol
2. On-page connectors are referenced using numbers
3. Off-page connectors are referenced using alphabets
4. General flow of processes is top to bottom or left to right
5. Arrows should not cross each other



Advantages:

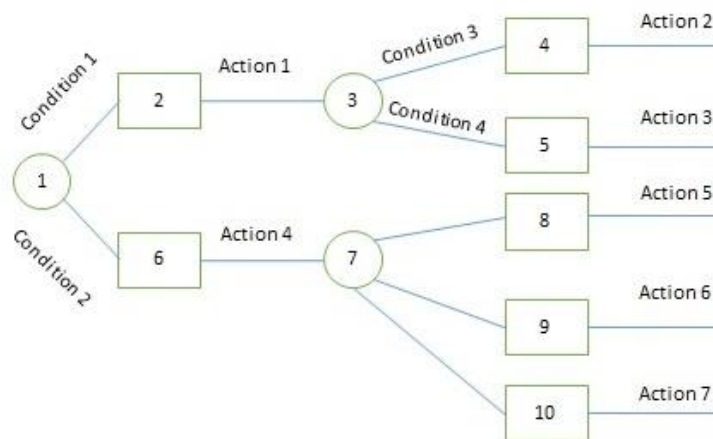
1. **Clarity and Communication:** Flowcharts provide a clear and visual representation of a process, making it easier for individuals to understand the sequence of steps involved.
2. **Problem-Solving:** They can serve as a tool for problem-solving by visualizing complex processes
3. **Documentation:** Flowcharts serve as documentation for processes, aiding in training new employees and providing a reference for established procedures.
4. **Process Improvement:** By visualizing processes, flowcharts facilitate brainstorming and collaboration, leading to more efficient process design and improvement initiatives.

Disadvantages:

1. Flowcharts can become overly complex, especially for processes with multiple decision points or branches, making them difficult to understand and interpret.
2. **Time-Consuming:** Developing detailed flowcharts can be time-consuming.
3. **Maintenance:** Flowcharts require regular maintenance to ensure they remain up-to-date with any changes to the process, which can be resource-intensive.

Decision Trees

- Decision tree is a tree like structure that represents the various conditions and the subsequent possible actions.
- It also shows the priority in which the conditions are to be tested or addressed.
- Each of its branches stands for any one of the logical alternatives and because of the branch structure, it is known as a tree.
- Decision trees are a method for defining complex relationships by describing decisions and avoiding the problems in communication.
- A decision tree is a diagram that shows alternative actions and conditions within horizontal tree framework. Thus, it depicts which conditions to consider first, second, and so on.
- Decision trees depict the relationship of each condition and their permissible actions.



Advantages:

- Simple to understand and to interpret. Trees can be visualize
- Requires little data preparation.
- Able to handle both numerical and categorical data.
- Able to handle multi-output problems.

Disadvantages:

- Decision trees can be unstable because small variations in the data might result in a completely different tree being generated.
- Decision tree learners create biased trees if some classes dominate.

Decision Tables

- Decision tables are a method of describing the complex logical relationship in a precise manner which is easily understandable.
- It is useful in situations where the resulting actions depend on the occurrence of one or several combinations of independent conditions.
- It is a matrix containing row or columns for defining a problem and the actions.

Components of a Decision Table

- **Condition Stub** – It is in the upper left quadrant which lists all the condition to be checked.
- **Action Stub** – It is in the lower left quadrant which outlines all the action to be carried out to meet such condition.
- **Condition Entry** – It is in upper right quadrant which provides answers to questions asked in condition stub quadrant.
- **Action Entry** – It is in lower right quadrant which indicates the appropriate action resulting from the answers to the conditions in the condition entry quadrant.
- For example, refer the following table –

CONDITIONS	Rule 1	Rule 2	Rule 3	Rule 4
Advance payment made	Y	N	N	N
Purchase amount = Rs 10,000/-	-	Y	Y	N
Regular Customer	-	Y	N	-
ACTIONS				
Give 5% discount	X	X	-	-
Give no discount	-	-	X	X

FEASIBILITY STUDY:-

- Feasibility Study is study to reveal whether project is feasible or not.
- It is conducted to find answer the following question:
 - do we have required resources and technologies to build a project
 - Do we receive profit from the project?
- It tells us whether project is worth the investment
- After feasibility study, project may be accepted, accepted with modification and rejected.
- Feasibility Study can be considered as preliminary investigation that helps the management to take decision about whether study of system should be feasible for development or not.
- A feasibility study is an assessment of the practicality and viability of a proposed project, venture, or business idea. It helps stakeholders determine whether the project is worth pursuing or not.

Types Of Feasibility study: -

- **Technical Feasibility Study:**
 - ✓ Technical feasibility studies evaluate whether the proposed project can be technically implemented.
 - ✓ This includes assessing factors such as the availability of technology, infrastructure, resources, and expertise needed to complete the project successfully.
- **Operational Feasibility**
 - ✓ It ensures that the management should support the proposed system and its working feasible in the current organizational environment.
 - ✓ It analyzes whether the users will be affected and they accept the modified or new business methods that affect the possible system benefits.
- **Legal and Regulatory Feasibility Study:**
 - ✓ Legal and regulatory feasibility studies examine the legal and regulatory requirements that may impact the project.
 - ✓ They identify any permits, licenses, compliance issues, or legal constraints that need to be addressed to ensure the project's success and compliance with laws and regulations.
- **Economic Feasibility**
 - ✓ The main aim of Economic Feasibility Analysis (EFS) is to estimate the economic requirements of candidate system before investments funds are committed to proposal.
- **Scheduling Feasibility Study:** Scheduling feasibility studies evaluate the project timeline and determine whether the proposed project can be completed within the desired timeframe.

• Importance of feasibility study: -

- Get clear cut idea whether the project is likely to be successful before allocating the budget manpower and time
- Improve project team focus

- Provides valuable information for go/no go decision
- Identify the valid reason to undertake the project.
- Enhance the success rate by evaluating multiple parameters

Steps Involved in Feasibility Analysis

The following steps are to be followed while performing feasibility analysis –s

- Form a project team and appoint a project leader.
- Develop system flowcharts.
- Identify the deficiencies of current system and set goals.
- Enumerate the alternative solution or potential candidate system to meet goals.
- Determine the feasibility of each alternative such as technical feasibility, operational feasibility, etc.
- Weight the performance and cost effectiveness of each candidate system.
- Rank the other alternatives and select the best candidate system.
- Prepare a system proposal of final project directive to management for approval.

